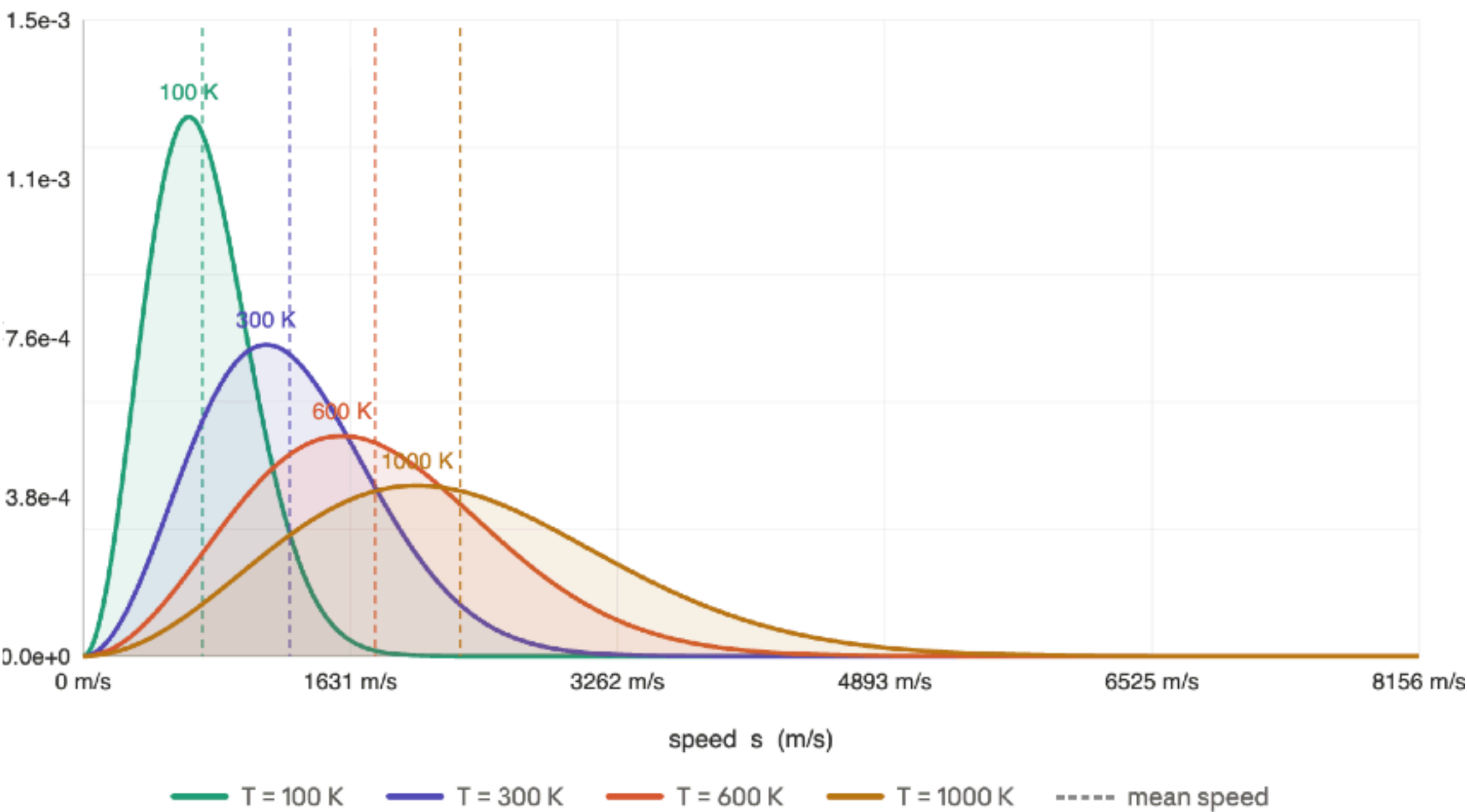




18

$$\pi_{\beta}(s) \propto s^2 \exp\left(-\frac{ms^2}{2k_B T}\right)$$





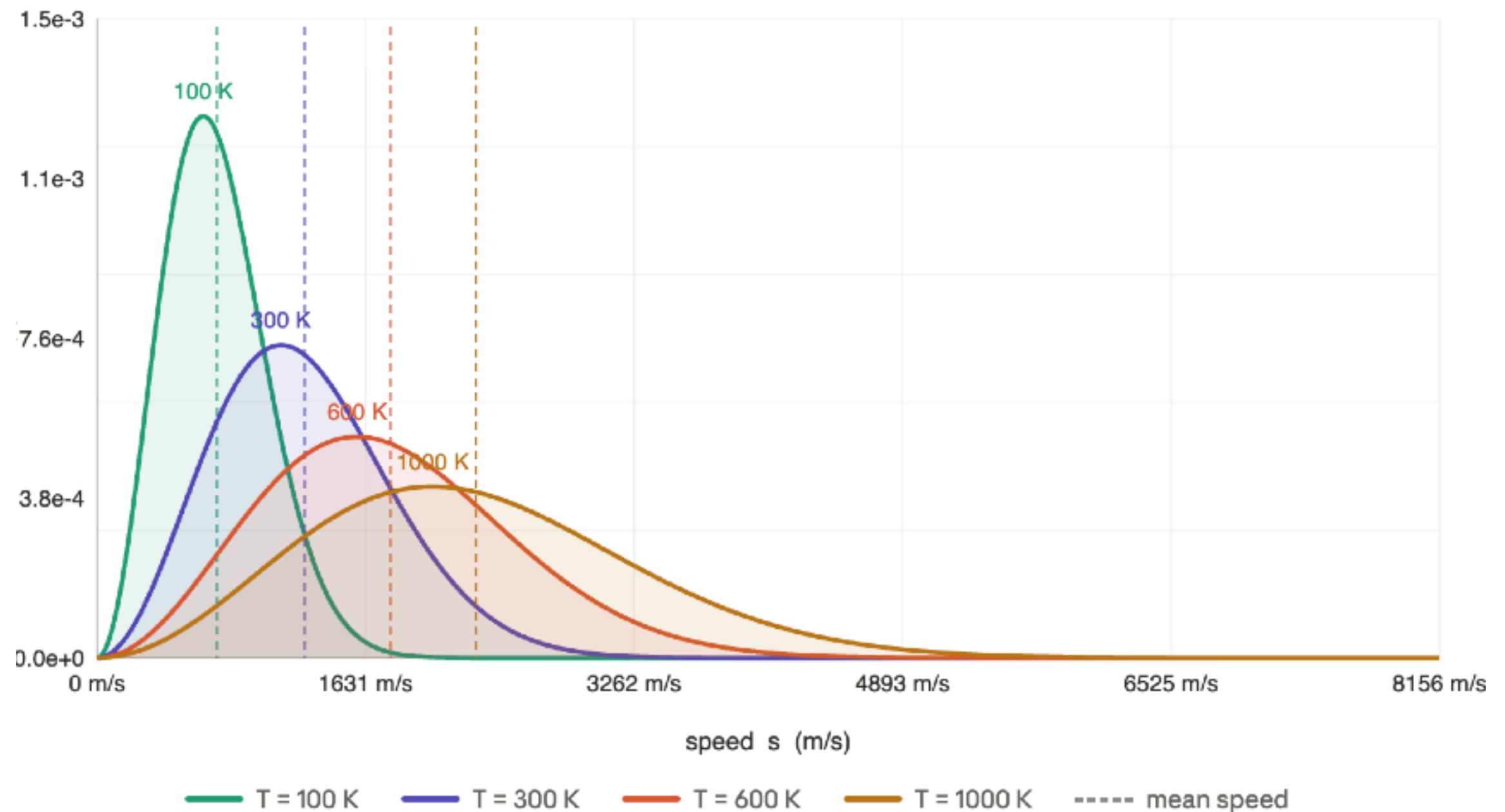
With $n = 4v$ we have

EXAMPLE: GAS IN A BOX

18

$$\pi_{\beta}(s) \propto s^2 \exp\left(-\frac{ms^2}{2k_B T}\right)$$

- With $m = 4u$ we have



EXAMPLE: ISING MODEL

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- ▶ Let $\mathbb{X} = \{-1, 1\}^V$ where V is the a d -dimensional lattice
- ▶ The energy forces neighbouring spins to prefer alignment, penalised by an external field

$$U(x) = -J \sum_{i \sim j} x_i x_j - h \sum_i x_i$$

- ▶ The first term favours aligned neighbours with coupling parameter J
- ▶ The second term favours all spins aligning with magnetic moment h

$$\pi_\beta(x) \propto \exp \left(\beta J \sum_{i \sim j} x_i x_j + \beta h \sum_i x_i \right)$$

- ▶ Total magnetisation of the state x is $m(x)$

$$m(x) = \sum_{i=1} x_i$$

